Final - Jie Wen

1

Let $c(K)$ be a 1Y European call payoff $(S_T - K)^+$. The specific payoff is given by:

$$\max\left((S_T - 50)^+, 2(S_T - 75)\right).$$

The payoff structure is:

$$\text{Payoff} = \begin{cases} 0, & S_T < 50 \\ S_T - 50, & 50 \le S_T < 100 \\ 2S_T - 150, & S_T \ge 100 \end{cases}$$

which is equivalent to:

$$(S_T - 50)^+ + (S_T - 100)^+ = c(50) + c(100).$$

Consequently, the traded price satisfies:

$$f_0 = C(50) + C(100).$$

The derivative is exactly a portfolio of two calls, so values and Greeks add linearly.

a)

The Black–Scholes formula with continuous dividend yield q is used:

$$C(S,K) = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2), \quad d_1 = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}},\ d_2 = d_1 - \sigma\sqrt{T}.$$

Given the parameters:

$$C(50; \sigma = 0.35) = 10.0247.$$

Therefore, the implied market price of the $K = 100$ call must be:

$$C(100) = f_0 - C(50) = 10.94 - 10.0247 = 0.9153.$$

Next, the implied volatility σ_{100} that satisfies $C(100) = 0.9153$ is calculated. Solving the Black-Scholes equation yields:

$$\sigma_{100} = 0.4030 \quad (40.3000\%).$$

Conclusion regarding the volatility structure:

- $\sigma_{\text{imp}}(K = 50, T = 1) = 35.0000\%$ (given)
- $\sigma_{\text{imp}}(K = 100, T = 1) \approx 40.3000\%$ (implied by the traded price)

At $T = 1$, the implied volatility is **higher at the higher strike** (indicating an upward-sloping right wing or "smile" on the call side). Therefore, the surface is **not flat** across strikes.

b)

For a European call under continuous dividend yield q:

$$\Delta_{\text{call}} = \frac{\partial C}{\partial S} = e^{-qT}N(d_1).$$

Since $f = C(50) + C(100)$, the deltas add linearly:

$$\Delta_f = \Delta_{50} + \Delta_{100} = 0.6627 + 0.1013 = 0.7640.$$

2

Let S_T be the stock price at maturity $T = 1$.

- If $S_T < 100$: $f_2 = 0$, $f_1 = 90 - S_T \Rightarrow f_2 - f_1 = S_T - 90$.
- If $S_T \geq 100$: $f_2 = S_T - 90$, $f_1 = 0 \Rightarrow f_2 - f_1 = S_T - 90$.

Thus, for all S_T,

$$f_2 = f_1 + (S_T - 90).$$

A claim that pays $S_T - K$ at T is the payoff of a long forward with delivery price K. With discrete dividends, the forward value at $t = 0$ is

$$\text{Value}(S_T - K) = S_0 - \text{PV}(D) - Ke^{-rT},$$

where $\text{PV}(D) = \sum_i D_i e^{-rt_i}$.

By $PV = CFe^{-rT}$ under continuous compounding:

$$\text{PV}(D) = 5e^{-0.0425 \cdot 0.5} \approx 4.8949,$$

$$90e^{-0.0425} \approx 86.2551.$$

$$\therefore \text{Value}(S_T - 90) = 95 - 4.8949 - 86.2551 = 3.8500.$$

$$\therefore V_0(f_2) = V_0(f_1) + \text{Value}(S_T - 90) = 8.7250 + 3.8500 = 12.5750.$$

3

Since the issuer **cannot default before maturity**, the **single-period (default-at-maturity) zero-coupon bond model** is used:

$$B(0,T) = N\Big(P_s(0,T) + R\big(1 - P_s(0,T)\big)\Big)e^{-rT} = N\Big(e^{-\lambda T} + R(1 - e^{-\lambda T})\Big)e^{-rT},$$

with $P_s(0,T) = e^{-\lambda T}$.

Given: $N = 1000, T = 2, r = 0.04, R_{\text{sen}} = 0.60, B_{\text{sen}} = 866.9$.

Then

$$866.9 = 1000\Big(e^{-\lambda T} + 0.6(1 - e^{-\lambda T})\Big)e^{-rT} = 1000(0.6 + 0.4e^{-\lambda T})e^{-0.08}.$$

$$\therefore e^{-\lambda T} \approx 0.8477539.$$

$$\therefore \lambda = -\frac{1}{T}\ln(0.8477539) \approx 0.0825825.$$

$$B_{\text{jun}} = 1000\Big(e^{-\lambda T} + 0.2(1 - e^{-\lambda T})\Big)e^{-0.08} = 1000(0.2 + 0.8e^{-\lambda T})e^{-0.08} = 810.6837.$$

4

a)

Under CAPM, each stock's **required (expected) return** is determined by its beta:

$$\mu_i = r + \beta_i(\mu_M - r)$$

The **1-period pricing/discounting** relationship (with $T = 1$) is:

$$S_i(0) = \frac{\mathbb{E}[S_i(1)]}{1 + (r + \beta_i(\mu_M - r))T}$$

$$\mu_M - r = 11\% - 4\% = 7\%$$

Stock A ($\beta_A = 0.25$):

$$\mu_A = 4\% + 0.25 \times 7\% = 5.75\%$$

Stock B ($\beta_B = 1.5$):

$$\mu_B = 4\% + 1.5 \times 7\% = 14.5\%$$

Using $S(0) = \frac{\mathbb{E}[S(1)]}{1 + \mu_i}$ (since $T = 1$):

Stock A ($\mathbb{E}[S_A(1)] = 49$):

$$S_A(0) = \frac{49}{1 + 0.0575} = 46.3357$$

Stock B ($\mathbb{E}[S_B(1)] = 118$):

$$S_B(0) = \frac{118}{1 + 0.1450} = 103.0568$$

b)

Market values today:

- Stock A position value: $V_A = (-10000) \times 50 = -500{,}000$
- Stock B position value: $V_B = (10000) \times 100 = 1{,}000{,}000$
- Total portfolio value: $V_P = V_A + V_B = 500{,}000$

The portfolio return is the weighted sum $R_P = \sum_i w_i R_i$ with $w_i = V_i / V_P$, and weights will be negative when short selling is allowed:

$$w_A = \frac{-500{,}000}{500{,}000} = -1, \qquad w_B = \frac{1{,}000{,}000}{500{,}000} = 2$$

Using the market model / CAPM regression form with uncorrelated idiosyncratic terms:

$$\mathrm{Cov}(R_A, R_B) = \beta_A \beta_B \mathrm{Var}(R_M)$$

(assuming the residuals across assets are uncorrelated).

$$\mathrm{Cov}(R_A, R_B) = 0.25 \times 1.5 \times 0.09 = 0.03375$$

For a two-asset portfolio,

$$\sigma_P^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \mathrm{Cov}(R_A, R_B) = 0.1225 + 0.81 - 0.135 = 0.7975$$

$$\sigma_P = \sqrt{0.7975} \approx 0.8930$$

c)

Under CAPM / the market model, **portfolio beta** is the value-weighted average of the constituents' betas (weights can be negative for short positions), and by the computing results in b):

$$\beta_P = w_A \beta_A + w_B \beta_B = (-1)(0.25) + (2)(1.5) = -0.25 + 3 = 2.75$$

d)

To achieve a **market neutral** portfolio (target $\beta^* = 0$) using an index future, the standard relationship for changing portfolio beta is used:

$$N = (\beta^* - \beta_P)\frac{P}{A}$$

where P is the **current portfolio value**, and A is the **value of the stocks (notional) underlying one futures contract**.
For equity index futures, the **contract notional** is the futures price times the contract multiplier. Given the contract value is 8800 per contract, $A = 8800$.

Using the given current prices (computed in question b)):

$$P = V_A + V_B = 500{,}000$$

$$\therefore N = (0 - 2.75)\frac{500{,}000}{8{,}800} = -2.75 \times 56.8181818 = -156.25$$

This indicates a **short** position in the index futures.

e)

The **parametric (normal) 1-day VaR** formula is used. For a short horizon (ignoring drift),

$$P(\Delta\Pi \le -VaR) = 1 - \alpha \Rightarrow VaR = -\sigma_\Pi \Pi_0 \sqrt{\Delta t} N^{-1}(1-\alpha),$$

and for $\alpha = 95\%$, $N^{-1}(1-\alpha) = N^{-1}(0.05) = -1.644853$.
With an annualization factor of 250, $\sqrt{\Delta t} = \sqrt{1/250}$.

Numerically:

$$VaR_{0.95,1d} = 1.644853 \cdot \sigma_\Pi \cdot \frac{1}{\sqrt{250}} \cdot \Pi_0.$$

Portfolio value today:

$$\Pi_0 = (-10000)(50) + (10000)(100) = 500{,}000.$$

1) VaR before adding futures

The portfolio annual return volatility is

$$\sigma_\Pi = 0.8930.$$

$$\therefore VaR_{\text{before}} = 1.644853 \cdot 0.8930 \cdot \frac{500{,}000}{\sqrt{250}} \approx \boxed{46450.7363}.$$

2) VaR after adding the index future hedge (market neutral)

After hedging to $\beta = 0$, the **systematic (market) variance** is removed, leaving only the idiosyncratic part. Using the one-factor decomposition

$$\sigma_\Pi^2 = \beta_\Pi^2 \sigma_M^2 + \sigma_{\epsilon,\Pi}^2,$$

with $\beta_\Pi = 2.75$ and $\sigma_M = 0.30$:

$$\sigma_{\epsilon,\Pi}^2 = 0.7975 - (2.75^2)(0.30^2) = 0.116875 \Rightarrow \sigma_{\epsilon,\Pi} = 0.3419.$$

The hedged (post-future) 1-day VaR is:

$$VaR_{\text{after}} = 1.644853 \cdot 0.3418699 \cdot \frac{500{,}000}{\sqrt{250}} = \boxed{17782.3056}.$$